* Chapter - 1 *

* Boolean Theorems and GATES *

* Boolean Theorems :-

1] Related to 'OR' Operation [+]

① $A + 0 = A$
$0 + A = A$
$\bar{A} + 0 = \bar{A}$
$AB + 0 = AB$

② $A + 1 = 1$
$\bar{A} + 1 = 1 \Rightarrow 1 + \text{anything} = 1$
$AB + 1 = 1$

③ $A + \bar{A} = \bar{A} + A = 1$
$AB + \overline{AB} = 1$

④ $A + A = A$
$A + A + A = A$
$\bar{A} + \bar{A} + \bar{A} = \bar{A}$
$AB + AB = AB$

⑤ $\overline{A \cdot B} = \bar{A} + \bar{B}$
$\overline{A + B} = \bar{A} \cdot \bar{B}$

2] Related to 'AND' Operation [.]

① $A \cdot 0 = 0$
$\bar{A} \cdot 0 = 0 \Rightarrow 0 \cdot \text{anything} = 0$
$AB \cdot 0 = 0$

② $A \cdot 1 = A$
$\bar{A} \cdot 1 = \bar{A}$
$AB \cdot 1 = AB$

③ $A \cdot \bar{A} = \bar{A} \cdot A = 0$
$AB \cdot \overline{AB} = 0$

④ $A \cdot A = A$
$\bar{A} \cdot \bar{A} = \bar{A}$
$AB \cdot AB = AB$
$A \cdot A \cdot A = A$

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3] Very imp Boolean Theorems

① $A.(B+C) = A.B + A.C$ → AND is Distributi over OR operation

② $A + (B.C) = (A+B).(A+C)$ → OR is distribut over AND

③ $AB + \bar{B} = (A.B) + \bar{B} = (A+\bar{B}).(B+\bar{B})$
$= (A+\bar{B}).1$
$= A + \bar{B}$

④ Common Theorem

- $(A+B).(A+C) = A + B.C$
- $(A+B)(B+C) = B + A.C$
- $(A+B+C)(A+\bar{B}) = A + (B+C).\bar{B}$
$= A + B.\bar{B} + \bar{B}.C$
$= A + 0 + \bar{B}.C$
$= (A+\bar{B}).(A+C)$
- $(A+B+C)(A+\bar{B}+C)(A+B+\bar{C})$
$= A + (B+C).(\bar{B}+C).(B+\bar{C})$

⑤ Redundancy Theorem :-

- $AB + \bar{A}C + BC = AB + \bar{A}C$

BC → Redundant Term

- $\bar{A}B + B\bar{C} + A\bar{C} = B\bar{C} + \bar{A}C$

Redundant Term

- $AB + \bar{B}c + \bar{A}c \rightarrow$ Not Applicable

⑥ Demorgan Theorem

- $\overline{A.B.C} = (\bar{A} + \bar{B} + \bar{C})$
- $\overline{A + B + C} = \bar{A} . \bar{B} . \bar{C}$
- $\overline{AB + BC} = \overline{AB} . \overline{BC}$
- $\overline{B.(A+C)} = \bar{B} + (\overline{A+C})$

$\bar{B} + \bar{A} . \bar{C}$

* SOP and POS form :-

SOP → Sum of Product

0 1

$\bar{A}$ A

POS → Product of Sum

0 1

A $\bar{A}$

	2^1 A	2^0 B	SOP Term	POS Term	F(A,B)
[0]	0	0	$\bar{A} . \bar{B}$	$A + B$	1
[1]	0	1	$\bar{A} . B$	$A + \bar{B}$	0
[2]	1	0	$A . \bar{B}$	$\bar{A} + B$	1
[3]	1	1	$A . B$	$\bar{A} + \bar{B}$	0

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$f(A,B) = \Sigma(0,2) = \bar{A}\bar{B} + A\bar{B}$

$= \bar{B} \cdot (A + \bar{A})$

$= \bar{B} \cdot 1 = \bar{B}$

$f(A,B) = \Pi(1,3) = (A+\bar{B}) \cdot (\bar{A}+\bar{B})$

$= \bar{B} + A \cdot \bar{A}$

$= \bar{B} + 0 = \bar{B}$

$f(A,B) = \underbrace{\Sigma(0,2)}_{\text{Minterm}} = \underbrace{\Pi(1,3)}_{\text{Maxterm}}$

- $f(A,B) = \Sigma(0,1,2,3) = \Pi(\text{Nil})$

$= \bar{A}\cdot\bar{B} + \bar{A}\cdot B + A\cdot\bar{B} + A\cdot B$

$= \bar{A} \cdot (B+\bar{B}) + A \cdot (B+\bar{B})$

$= \bar{A} \cdot 1 + A \cdot 1$

$= A + \bar{A}$

$= 1 \rightarrow$ Always

$f(A,B) = \Pi(0,1,2,3) = \Sigma(\text{Nil})$

$= (A+B) \cdot (A+\bar{B}) \cdot (\bar{A}+B) \cdot (\bar{A}+\bar{B})$

$= [A + B\cdot\bar{B}] \cdot [\bar{A} + B\cdot\bar{B}]$

$= A \cdot \bar{A}$

$= 0 \rightarrow$ Always

* GATES :-

- Basic Gates → NOT, AND, OR
- Arithmetic Gates → XOR, XNOR
- Universal Gates → NAND, NOR

1] NOT Gate :-

Representation → $A \rightarrow \bar{A}$ (Inverting Buffer), $A \rightarrow A$ (Buffer)

Truth Table

A	o/p
0	1
1	0

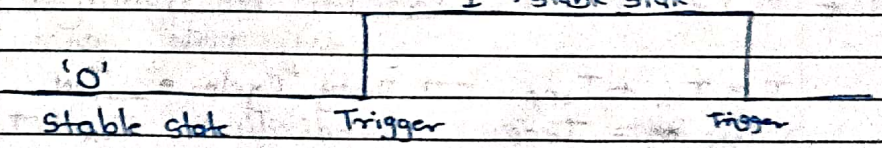
$\Rightarrow y(A) = \Sigma(0) = \Pi(1)$

$= \bar{A} \quad = \bar{A}$

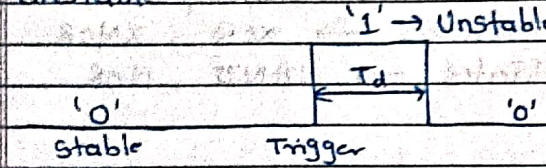
- Multivibrator :-

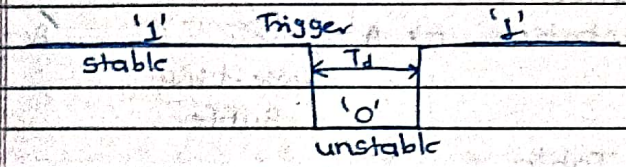
① Bistable Multivibrator

– Both states [state 0 and state 1] are stable state

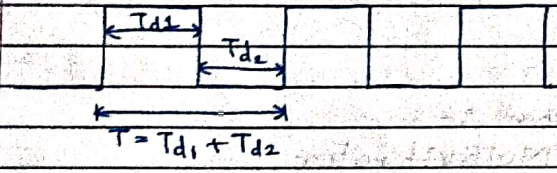


② ~~Bistable~~ Monostable Multivibrator
- One state is stable and other state is unstable





③ Astable Multivibrator → Square waveform Generator
- Both states are unstable states





$T = T_{ON} + T_{OFF}$ → Duty Cycle = $\frac{T_{ON}}{T_{ON} + T_{OFF}}$

$T_{ON} = T_{OFF}$

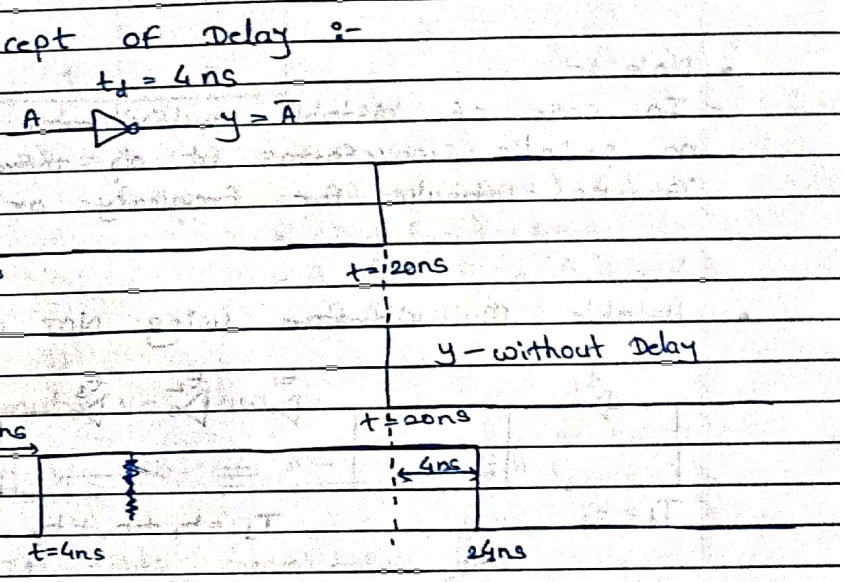
% Duty Cycle = $\frac{T_{ON}}{T_{ON} + T_{OFF}} \times 100\ \%$

① If $T_{ON} = T_{OFF}$ → Duty Cycle = 50%

② If $T_{ON} > T_{OFF}$ → Duty Cycle > 50%

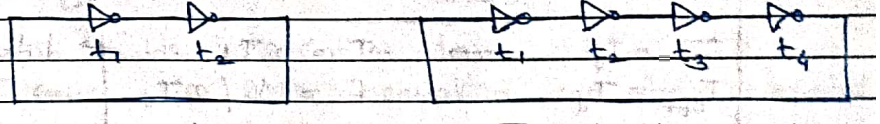
③ If $T_{ON} < T_{OFF}$ → Duty Cycle < 50%

* Concept of Delay :-



* Designing of Astable and Bistable Multivibrator using NOT gate

• Bistable multivibrator using NOT Gate



$T_d = t_1 + t_2$

$T_d = t_1 + t_2 + t_3 + t_4$

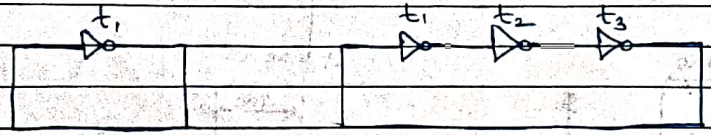
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- When even no of NOT gates are cascaded and o/p of last NOT gate is feedback to the i/p of first NOT gate then we get bistable multivibrator

- Buffer operation in a loop is equivalent to Bistable multivibrator

• Note :-
- In case of bistable multivibrator there is no periodic waveform at o/p therefore there is no formula for frequency or time period

• Astable Multivibrator using NOT Gate



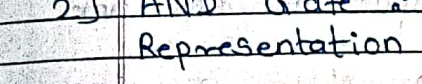
$T_d = t_1$ $\quad$ $T_d = t_1 + t_2 + t_3$

- When odd no. of NOT gates are connected in cascade and o/p of last NOT gate is feedback to the i/p of first NOT gate, then we get Astable multivibrator

- Inverter operation in a loop is equivalent to Astable Multivibrator.

$T_{ON} = T_d$, $T_{OFF} = T_d$ $\longrightarrow$ $T = 2T_d$, $f = 1/2T_d$ $\rightarrow$ duty cycle $\Rightarrow$ 50%

2] AND Gate :-

Representation $\longrightarrow$ 

$A.B = y$

- Truth Table

	A	B	y = A.B
0	0	0	0
1	0	1	0
2	1	0	0
3	1	1	1

$y(A,B) = \Sigma(3) = \Pi(0,1,2)$

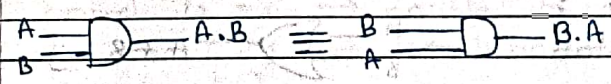
$= A.B = (A+B).(A+\bar{B}).(\bar{A}+B)$

$= (A + B.\bar{B})(\bar{A}+B)$

$= A.(\bar{A}+B) = A.\bar{A} + A.B$

$= A.B$

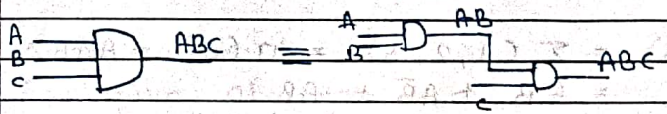
- Commutative law :-



- It holds commutative law
- Position of variables are irrelevant

$ABC = ACB = CAB = BAC = BCA$

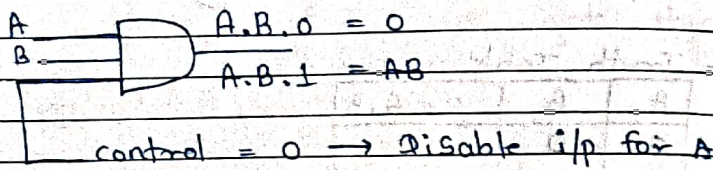
- Associative law



- It holds associative law
- Multi i/p AND operation can be generated using 2-i/p AND operation gates

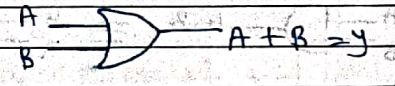
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- Enable and Disable input for AND Gate



control = 0 → Disable i/p for AND gate
= 1 → enable i/p for AND gate

- Imp Points :-
- If any one of the i/p line is at logic '0' then irrespective of other i/p lines o/p has to be '0'.
- o/p will be '1' only if all the i/p lines together are at logic '1'.
- Disable i/p = 0
Enable i/p = 1

3] OR Gate :-
Representation → 

	A	B	y = A+B
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1

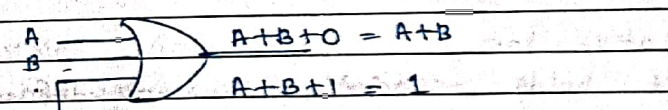
$y(A,B) = \Sigma(1,2,3) = \Pi \cdot (0) = A+B$
$= \bar{A}B + A\bar{B} + AB$
$= A + \bar{A} \cdot B$
$= (A+\bar{A}) \cdot (A+B)$
$= (A+B)$

- Commutative law :-
$A+B = B+A$
- It holds commutative law
- position of variables is irrelevant

$A+B+C = B+A+C = B+C+A = C+B+A = C+A+B$

- Associative law :-
- It holds Associative law
- It means we can generate multi i/p OR operation using 2-i/p OR gates

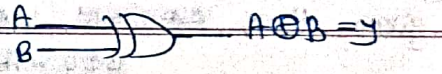
- Enable and Disable input for 'OR' gate



control i/p = 0 → enable i/p for OR gate
= 1 → disable i/p for OR gate

- Imp Points
- If any of i/p line is at logic '1' then irrespective of other i/p lines o/p will be at logic '1'
- o/p will be '0' only if all the i/p lines together will be at logic '0'.

4] XOR - Gate :-
- Exclusive OR gate
- It is defined for two i/p only
- Representation :-



- Truth Table

	A	B	y = A⊕B
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	0

$y(A,B) = \Sigma(1,2) = \Pi(0,3)$

$\Sigma(1,2) = \bar{A}B + A\bar{B}$

$\Pi(0,3) = (A+B).(\bar{A}+\bar{B})$

$A \oplus A = \bar{A}.B + A.\bar{B}$

- Commutative law
  - It holds commutative Law
  - Position of variables is irrelevent
  - $A \oplus B = B \oplus A$

- Associative law
  - It holds associative law → defined to calculated multi i/p XOR operation

• Properties of XOR Gate :-

① $A \oplus A = \bar{A}.A + A.\bar{A} = 0$

$\bar{A} \oplus \bar{A} = 0$

$AB \oplus AB = 0$

② $A \oplus \bar{A} = \bar{A}.\bar{A} + A.\bar{\bar{A}} = 1$

$AB \oplus \overline{AB} = 1$

④ $A \oplus \bar{A} = 1 \leftrightarrow A \oplus 1 = \bar{A}$

$\bar{A} \oplus 1 = \bar{\bar{A}} = A$

$AB \oplus 1 = \overline{AB}$

③ $A \oplus 0 = A$

$\bar{A} \oplus 0 = \bar{A}$

$AB \oplus 0 = AB$

⑤ Exchang property

If, $A \oplus B = C$ given, then, $A \oplus C = B$, $C \oplus B = A$ &

$B \oplus C = A$

⑥ $A \oplus A \oplus A \oplus A \ldots\ldots n$ times [n = represents no. of A]

then o/p = 0 for n = even

= A for n = odd

⑦ Buffer and inverting buffer using XOR



$A \oplus 0 = A$ → act as buffer

$A \oplus 1 = \bar{A}$ → act as inverting buffer

→ controlling i/p = 0 → act as buffer
= 1 → act as inverting buffer

} both are erabl i/ps

⑤ XNOR Gate :-

- Exclusive NOR Gate
- It is also defined for two i/ps only

Representation :-



$A \odot B = y$

$A \odot B = \bar{A}\bar{B} + AB$

- Truth Table

	A	B	y = A⊙B
0	0	0	1
1	0	1	0
2	1	0	0
3	1	1	1

$y = (A,B) = \Sigma(0,3) = \Pi(1,$

$\Sigma(0,3) = \bar{A}\bar{B} + AB$

$\Pi(1,2) = (A+\bar{B}).(\bar{A}+B)$

$A \odot B = \bar{A}\bar{B} + AB$

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- Commutative law :-
  - It holds commutative law
  - position of variables is irrevelent
  - $A \odot B = B \odot A$

- Associative law
  - It holds Associative Law
  - Defined to calculate multi i/p XNOR

• Properties of XNOR Gate :-

① $A \odot A = \bar{A}.\bar{A} + A.A = 1$
$\bar{A} \odot \bar{A} = 1$
$AB \odot AB = 1$

② $A \odot \bar{A} = \bar{A}.\bar{\bar{A}} + A.\bar{A} = 0$
$AB \odot \bar{AB} = 0$

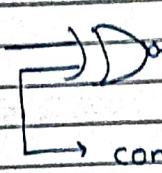
③ $A \odot A = 1 \rightarrow A \odot 1 = A$
$\bar{A} \odot 1 = \bar{A}$
$AB \odot 1 = AB$

④ $A \odot \bar{A} = 0 \rightarrow A \odot 0 = \bar{A}$
$\bar{A} \odot 0 = \bar{\bar{A}} = A$
$AB \odot 0 = \overline{AB}$

⑤ Exchange properties of XNOR Gate
If $A \odot B = C$ given, then,
$A \odot C = B$
$C \odot B = A$
$B \odot C = A$

⑥ $A \odot A \odot A \odot A \ldots\ldots$ n times [n - represents no. then o/p = 1 for n = even or A]
= A for n = odd

⑦ Buffer and inverting buffer using XNOR :-



$A \odot 0 = \bar{A}$ → inverting buffer
$A \odot 1 = A$ → buffer

→ controlling i/p = 0 → Act as inverting buff
= 1 → Act as buffer

0 and 1 both are enable i/ps.

* Combined Imp Properties of XOR and XNOR

• $\bar{A} \oplus B = \bar{\bar{A}}.B + \bar{A}.\bar{B} = A \odot B$
$A \oplus \bar{B} = \bar{A}.\bar{B} + A.\bar{\bar{B}} = A \odot B$
$\bar{A} \oplus \bar{B} = \bar{\bar{A}}.\bar{B} + \bar{A}.\bar{\bar{B}} = A \oplus B$
$\bar{A} \odot B = \bar{\bar{A}}.\bar{B} + \bar{A}.B = A \oplus B$
$A \odot \bar{B} = \bar{A}.\bar{\bar{B}} + A.\bar{B} = A \oplus B$
$\bar{A} \odot \bar{B} = \bar{\bar{A}}.\bar{\bar{B}} + \bar{A}.\bar{B} = A \odot B$

• $A \oplus B = \overline{A \odot B}$
$A \odot B = \overline{A \oplus B}$

• $A \oplus B \oplus C \neq \overline{A \odot B \odot C}$
$A \oplus B \oplus C = A \odot B \odot C$

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- XOR gate search for →
  Always odd no. of total 1's on i/p side
  ↳ i.e. if total no of 1's on i/p side are odd in number then o/p is '1' otherwise o/p is '0'

- XNOR gate search for →

① If i/p lines are even in no. then XNOR gate search for even no. of 1's
  ↳ i.e. if total no of 1's on i/p side is even in number then o/p is '1' otherwise o/p is '0'

② If i/p lines are odd in no. then XNOR gate search for odd no. of 1's
  ↳ i.e. if total no. of 1's on i/p side are odd in number then o/p is '1' otherwise o/p is '0'.

- Therefore for odd no of i/p lines
  XOR = XNOR
  $A \oplus B \oplus C = A \odot B \odot C$
  $A \oplus B \oplus C \oplus D \oplus E = A \odot B \odot C \odot D \odot E$
- for even no of i/p lines →
  XOR = $\overline{\text{XNOR}}$ or XNOR = $\overline{\text{XOR}}$
  $A \oplus B \oplus C \oplus D = \overline{A \odot B \odot C \odot D}$

- Compliment Property of XOR and XNOR Gate

① $\overline{A \oplus B} = \bar{A} \oplus B = A \oplus \bar{B} = A \odot B$
  $\overline{A \odot B} = \bar{A} \odot B = A \odot \bar{B} = A \oplus B$

② $\overline{A \oplus B \oplus C} = \bar{A} \oplus B \oplus C = A \oplus \bar{B} \oplus C = A \oplus B \oplus \bar{C}$
  $= A \odot B \oplus C = A \oplus B \odot C$

③ $\overline{A \oplus B \oplus C} = \overline{A \odot B \oplus C} = \bar{A} \odot B \oplus C = A \odot \bar{B} \oplus C$
  $= A \odot B \oplus \bar{C}$
  $\overline{\overline{A \oplus B \oplus C}} = A \oplus B \oplus C = A \odot B \odot C$

④ For even no of i/p lines XOR is ~~same~~ equal to compliment of XNOR or vice versa

⑤ For odd no. of i/p lines XOR is equal to XNOR

Imp ⑥ → Even compliment = zero compliment
  Odd compliment = One compliment

eg. ① $\overline{A \oplus B \odot \bar{C} \oplus \bar{D}}$
→ Here total compliments are odd = 5
so
  $\overline{A \oplus B \odot \bar{C} \oplus D}$
  $= A \oplus B \oplus C \oplus D$

② $\overline{\bar{A} \oplus \bar{B} \odot \bar{C} \odot \bar{D}}$
→ Here total compliment are 7
so
  $\overline{A \oplus B \odot C \odot D}$
  $= A \odot B \odot C \odot D \quad \overline{A \oplus B \oplus C \oplus D}$

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- Note :- XOR and XNOR does not hold distributive law over 'OR' operation or over 'AND' operation

Que : $A \oplus B \oplus \bar{A}B$ is equal to

→ we take two compliments to balance

$\overline{\bar{A} \oplus B \oplus \bar{A}B}$

No it is above form $x \oplus y \oplus xy \rightarrow x+y$

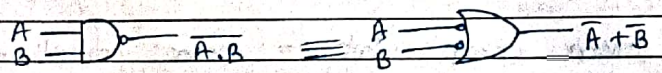
$x \oplus y \oplus (x+y) \rightarrow xy$

$= \overline{\bar{A} + B}$

$= A \cdot \bar{B}$

6] NAND GATE :- [AND + NOT]

Representation :-



A	B	$y = \overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

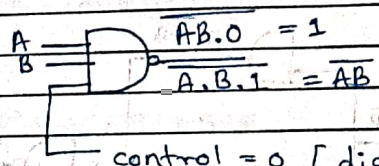
$y(A,B) = \sum(0,1,2)$

$= \pi(3)$

- Commutative law :-
  - It holds commutative law
  - position of variables are irrelevent
  - $\overline{A.B.C} = \overline{B.A.C} = \overline{B.C.A} = \overline{C.B.A}$

- Associative law :-
  - It does not hold associative law

- Imp points :-
  - If any one of the i/p line is at logic '0' then irrespective of other i/p lines o/p will be at logic '1'
  - o/p will be '0' only all the i/p together are at logic '1'



control = 0 [disable i/p for NAND gate]

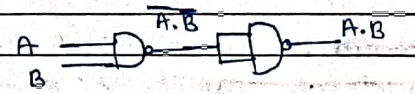
= 1 [enable i/p for NAND]

- NAND as a universal gate :-

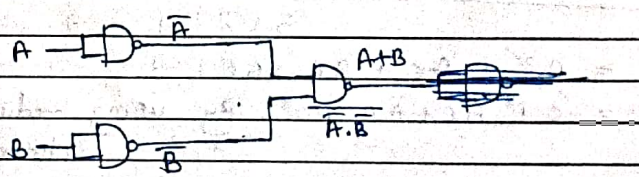
① NOT Gate



② AND Gate :-

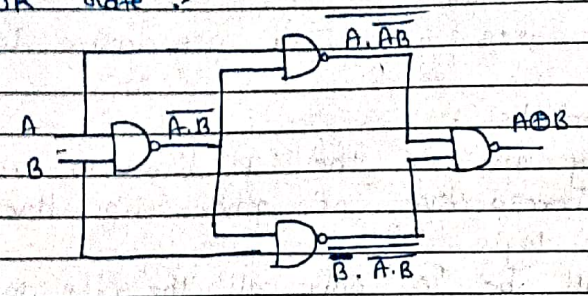


③ OR Gate :- A+B

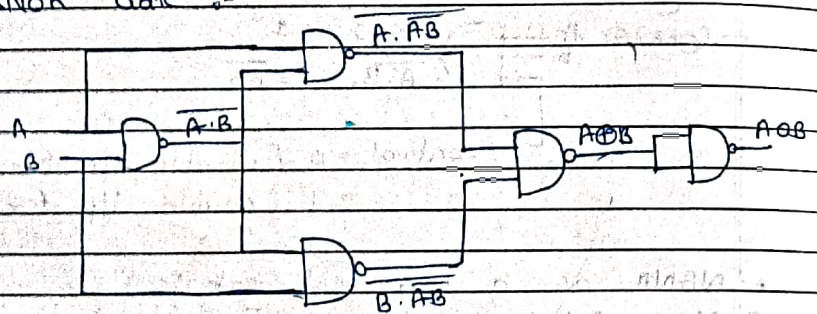


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- XOR Gate :-



- XNOR Gate :-



* How to solve the 2-input NAND Gate are minimum required.

① Transpose Theorem :-

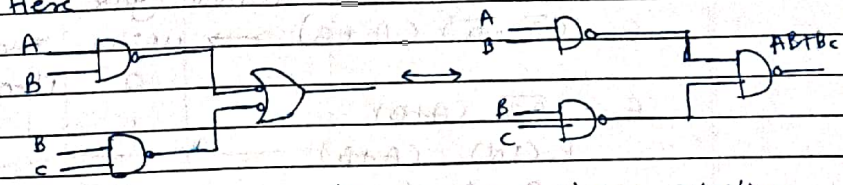
$$(A+B)(\bar{A}+C) = AC+\bar{A}B$$

$= A.\bar{A} + AC + \bar{A}B + BC$

$= 0 + AC + \bar{A}B$ .... By using redundancy Theorem

② $y = AB + BC$

Here



- first make the connection using given question then replace all gates by NAND Gate. But this way is not always correct sometimes it gives minimum NAND gates and sometimes not.

③ $f(A,B,C) = AB + BC$

- If given question is in OR form then use.

→ $\bar{f} = \overline{AB+BC}$

$\bar{f} = \bar{A}\bar{B} \cdot \bar{B}\bar{C}$

$\bar{f} = P(1N) \; Q(1N)$ ⇒ Total ⇒ 3 Nand gates

$\bar{f} = P.Q$

$f = \overline{P.Q}$

$\in 1(N)$

④ $f(A,B,C) = A\bar{B} + A\bar{C}$

- If • in given question hidden NAND Gate is present then use below method.

$f = A(\bar{B}+\bar{C})$

$= A(\overline{B.C})$

$P(1N)$ ⇒ Total = 3 Nand gate

$f = A.P$

$(2\in N)$

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④ $F_1(A,B) = \bar{A}B + A\bar{B}$

Here also hidden NAND gate is present

$F_1 = (\bar{A}+\bar{B})(A+B)$ → using Transpose Theorem

$F_1 = \overline{\bar{A}\bar{B}} \cdot (A+B)$

$P.(1N) \cdot (A+B)$

$= P.(A+B)$

$F_1 = P.A + P.B$ ⇒ Total = 4 NAND gates required

$\bar{F}_1 = \overline{P.A + P.B}$

$= \overline{P.A} \cdot \overline{P.B}$

$= R(1N) \rightarrow S(1N)$

$\bar{F}_1 = R.S$

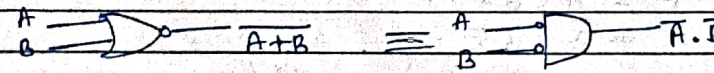
$F_1 = \overline{R.S}$

$(1N)$

- Note : ① If OR operation is comming and around that both side have atleast one term with compliment then there is always hidden NAND.
  ② Solving that hidden NAND problem there is above two ways
  ③ If question is not in form of hidden NAND then follow 5th F method

7) NOR Gate :-

Representation :-



A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

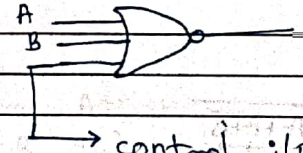
$y(A,B) = \Sigma(0)$

$= \pi(1,2,3)$

- Commutative law :-
  - It holds commutative law
  - Position of variables is irrelevant
  - $\overline{A+B+C} = \overline{B+A+C} = \overline{B+C+A} = \overline{C+B+A}$

- Associative law :-
  - It does not hold Associative law

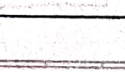
- Imp points :-
  - If any one of the i/p line is at logic '1' then irrespective of other i/p lines o/p will be at logic '0'.
  - O/p will be at logic '1' only all the i/p lines together are at logic '0'.



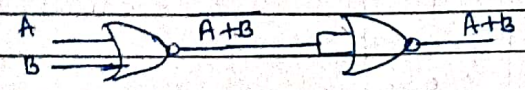
→ control i/p = 0 → enable i/p for NOR ga
= 1 → disable i/p for NOR ga

- NOR Gate as Universal Gate

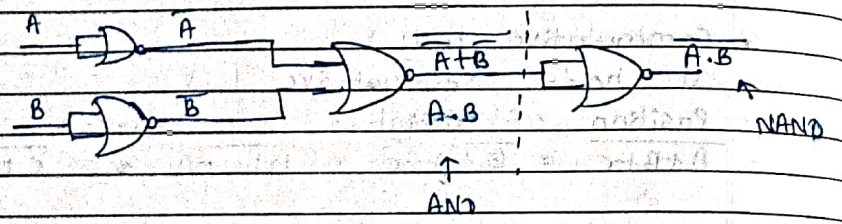
① NOT Gate



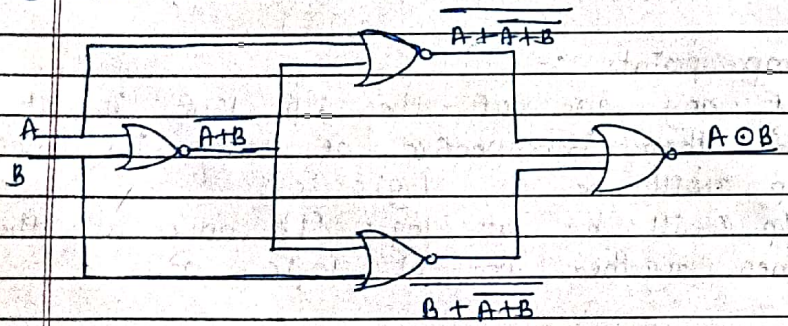
② OR Gate (A+B)



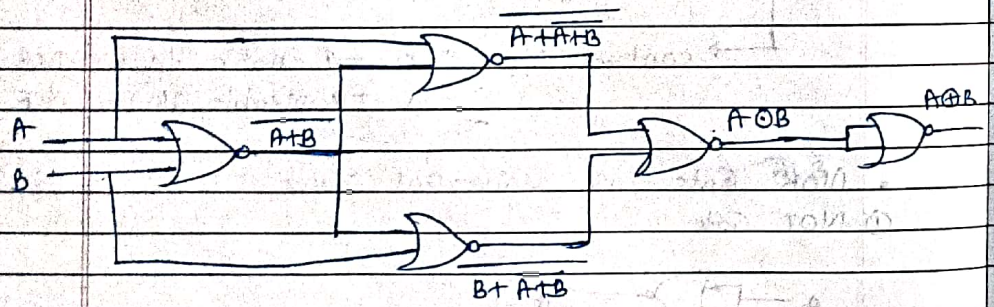
③ AND Gate (A.B)



④ XOR Gate

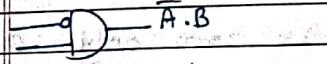


⑤

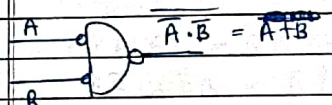


Gates	No. of NAND	No. of NOR
NOT	1	1
AND	2	3
OR	3	2
XOR	4	5
XNOR	5	4

• AND + NOT

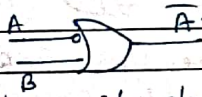


Here in o/p Not and AND is present so above gate is universal gate

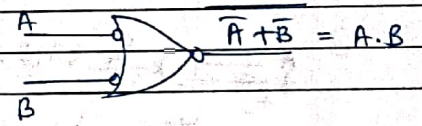


Here only OR is comming so above gate is not universal.

• OR + NOT



Here in o/p NOT and OR is present so above gate is universal gate



Here only AND is comming so above gate is not universal

→ Same As :—

(XOR, OR) → Universal → A⊕B⊕(A+B) = AB

(XOR, AND) → Universal → A⊕B⊕AB = A+B

(XNOR, OR) → Universal → A⊙B⊙(A+B) = A.B

(XNOR, AND) → Universal → A⊙B⊙(A.B) = A+B

XOR → NON - Universal

XNOR → Non - Universal

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- This is new Type of Question

- $f_1(A,B,C) = \Sigma(1,2,3,6,7)$
  $f_2(A,B,C) = \Sigma(0,2,3,5)$

⊙ Here above are also given in POS format but remember first of all convert both into SOP form.

① f_1, f_2 → XOR → $f_3(A,B,C)$

→ Here in XOR output is zero only if both the i/ps are same so in f_1 and f_2 2, 3 are present in both and 4 is absent in both so,

$f_3(A,B,C) = \Sigma(0,1,5,6,7) = \Pi(2,3,4)$

② A, B → NAND → $f_4(A,B,C)$; f_1, f_2

Here in NAND gate if both inputs are present then output will be zero so,

$f_4(A,B,C) = \Sigma(0,1,4,5,6,7) = \Pi(2,3)$

③ f_1, f_2 → NOR → $f_5(A,B,C)$ =

$= \Sigma(4) = \Pi(0,1,2,3,5,6,7)$

④ f_1, f_2 → XNOR → $f_6(A,B,C)$

$= \Sigma(0,1,2,3,5,6,7) = \Pi(4)$

* Concept of Duality :-

- How to find out Dual Boolean function of a given boolean function

  → Replace all OR operation (+) by AND (·)
  → Replace all AND operation (·) by OR (+)
  → NOT remains as it is. (NOT is replace by NOT)

By doing above steps we get dual boolean function of a given boolean function

eg. ① $f(A,B,C) = AB + BC = \Sigma(3,6,7)$
$f^D(A,B,C) = (A+B).(B+C)$
$= B + AC$

$f^D \neq f$
∴ It is Non Self-Dual boolean function

② $f(A,B) = \bar{A}\bar{B} + \bar{A}B$
$\bar{A}\bar{B} + \bar{A}B = \bar{A}[\bar{B}+B] = \bar{A}.1 = \bar{A} = \Sigma(0,1)$

$f^D(A,B) = (\bar{A}+\bar{B}).(\bar{A}+B)$
$= \bar{A} + \bar{B}.B$
$= \bar{A}$

$F = F^D$
∴ It is self dual boolean function

- Note :- So if for a given boolean function $F^D = F$ then we say that given boolean function is self dual boolean function otherwise if $F^D \neq F$ then non-self dual boolean function

② for any boolean function $(F^D)^D = F$
$F \xrightarrow{\text{Dual}} F^D \xrightarrow{\text{Dual}} F$

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① If F is self dual boolean function then $\bar{F}$ guaranteed to be self dual.

* Some Dual function or Gate

① AND $\xrightarrow{\text{Dual}}$ OR (A+B)
(A.B) $\xleftarrow{\text{Dual}}$

② OR $\xrightarrow{\text{Dual}}$ AND
$\xleftarrow{\text{Dual}}$

③ NAND $\xrightarrow{\text{Dual}}$ NOR
$\xleftarrow{\text{Dual}}$

④ XOR $\xrightarrow{\text{Dual}}$ XNOR
$\xleftarrow{\text{Dual}}$

⑤ NOT $\xrightarrow{\text{Dual}}$ NOT

⑥ 1 $\xrightarrow{\text{Dual}}$ 0
$\xleftarrow{\text{Dual}}$

* • n → variables
How many boolean function are possible to making

$N = 2^n$ → term (combination) [0 to $(2^n - 1)$]
if n = 3 variable, → $N = 2^3$ = [0,1,2,3,4,5,6,7]

$M = 2^N = 2^{2^n}$ → M = max Boolean funct^n making
$M = 2^N = 2^{2^3} = 2^8 = 256$ boolean funct^n.

So If 3 variables are present then total combinations will 8 $(2^n = 2^3)$ and total boolean functions are M = 256 $[M = 2^N = 2^8]$

And Now, How many boolean function are self dual in 256 functions are calculate using..

$SD = 2^{2^{n-1}} = 2^{2^{3-1}} = 2^{2^2} = 2^4 = 16$

• To find the function self dual or not is given below,
– Condition :→ ⊕
① Necessary condition :- Exactly Half of the term should be there in SOP or POS form
② The groups are divided like [for 3 combinatio
→ (0,7), (1,6), (2,5), (3,4)
this is for three variably
(first, last), (second, second last), (third, third last)...
③ So, the groups are containg only 2 = combin and in so for dual function in SOP form the values are same that must be only one from each group.

eg. $f(A,B) = \bar{A}B + A\bar{B} = \Sigma(1,2) = \Pi(0,3)$
So here for two variables 1 and 2 are in same group and both are comming in SOP form so above function is Non-self dual

• n = variable :-
$N = 2^n$ terms

groups $= \frac{N}{2} = \frac{2^n}{2} = (2^{n-1})$ group of two terms

$SD = 2^{2^n - 1}$

→ 4-variable → $N = 2^4 = 16$ terms
$M = 2^{16} = 64 \times 1024 = 65536$
$SD = 2^{2^4 - 1} = 2^{2^3} = 2^8 = 256$
$NSD = M - SD = 65536 - 256$
$= 65280$

* Canonical and Non-Canonical Form

– In given function if any one variable is missing then that is non-canonical and all variables are present then canonical

eg $F(A,B,C) = \bar{A}BC + A\bar{C} + \bar{B}C$
Here B and A are missing so non-canonical form

$F(A,B,C) = A\bar{B}C + \bar{A}BC + \bar{A}B\bar{C}$
Here all variables are presents so canonical form

• How to convert Non-canonical to canonical

$F(A,B,C) = AB + BC$
$= AB(C+\bar{C}) + (A+\bar{A})BC$
$= ABC + AB\bar{C} + ABC + \bar{A}BC$
$= ABC + AB\bar{C} + \bar{A}BC$
(111) (110) (011)
(7) (6) (3)
$= \Sigma(3,6,7)$

– Short method

eg $F(A,B,C) = AB + B\bar{C} + \bar{A}C$

$AB'C$	$AB\bar{C}$	$\bar{A}BC$
110 → 6	010 → 2	001 → 1
111 → 7	110 → 6	011 → 3

$= \Sigma(1,2,3,6,7)$
$= \Pi(0,4,5)$

* Radix compliment (r's complement)

r's compliment → $r^n - N$ → r = base, n = no. of digits, N = given no.

* Diminished Radix compliment

(r-1)'s compliment → $r^n - N - 1$

r' comp − 1

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